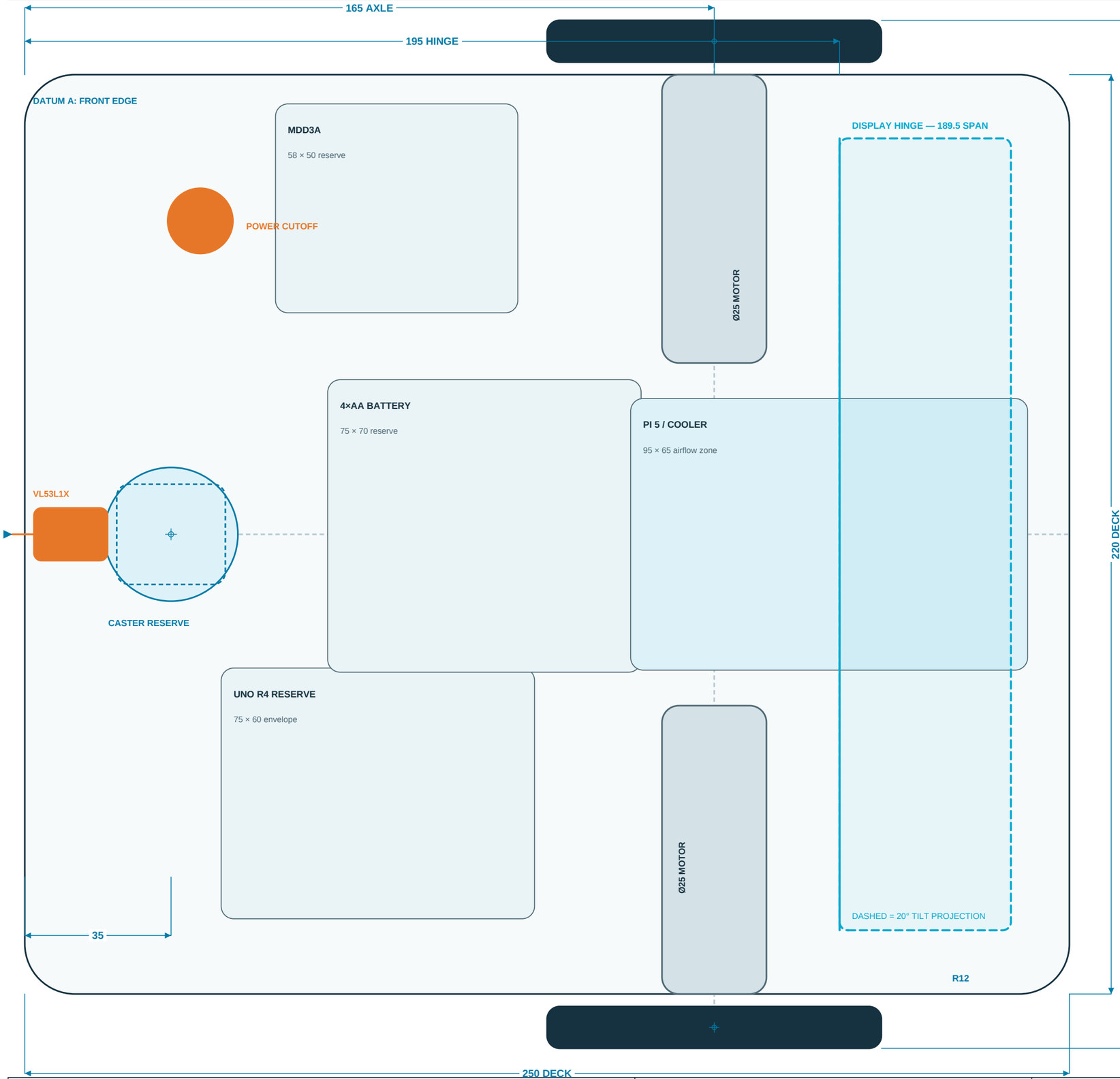
# GENERAL ARRANGEMENT — TOP VIEW

Controlled layout dimensions and reserved component envelopes






## CONTROLLED DIMENSIONS

Deck: 250 L × 220 W × 5 thick
Corner radius: R12
Drive axle: X = 165 from front datum
Caster center: X = 35, Y = 110
Display hinge: X = 195; centered on Y
Nominal wheel envelope: 246 overall

## LAYOUT RULES

Battery low and centered, ahead of axle.
Keep Pi cooler airflow zone unobstructed.
Use slots for caster and motor brackets.
Route motor wiring away from EEG leads.
All electronics on insulating standoffs.
Do not drill from this sheet until verified.

## DATUM / COORDINATE SYSTEM

Origin: front-left deck corner.
+X runs front → rear; +Y left → right.
Centerline: Y = 110.
Bottom face of deck is datum Z = 22.
Dimensions are nominal millimetres.
Drawing scale: 1:1 on A3 when printed 100%.

## REV A STATUS

Layout-ready, not hole-ready.
Measure the actual caster, brackets and display mounts before the final print.

| MUSE BOT BASE — MANUFACTURING SKETCH<br>Planning drawing; verify purchased hardware before production | DRAWING<br>MB-BASE-001 | REV<br>A | UNITS<br>mm | SCALE<br>1:1 |
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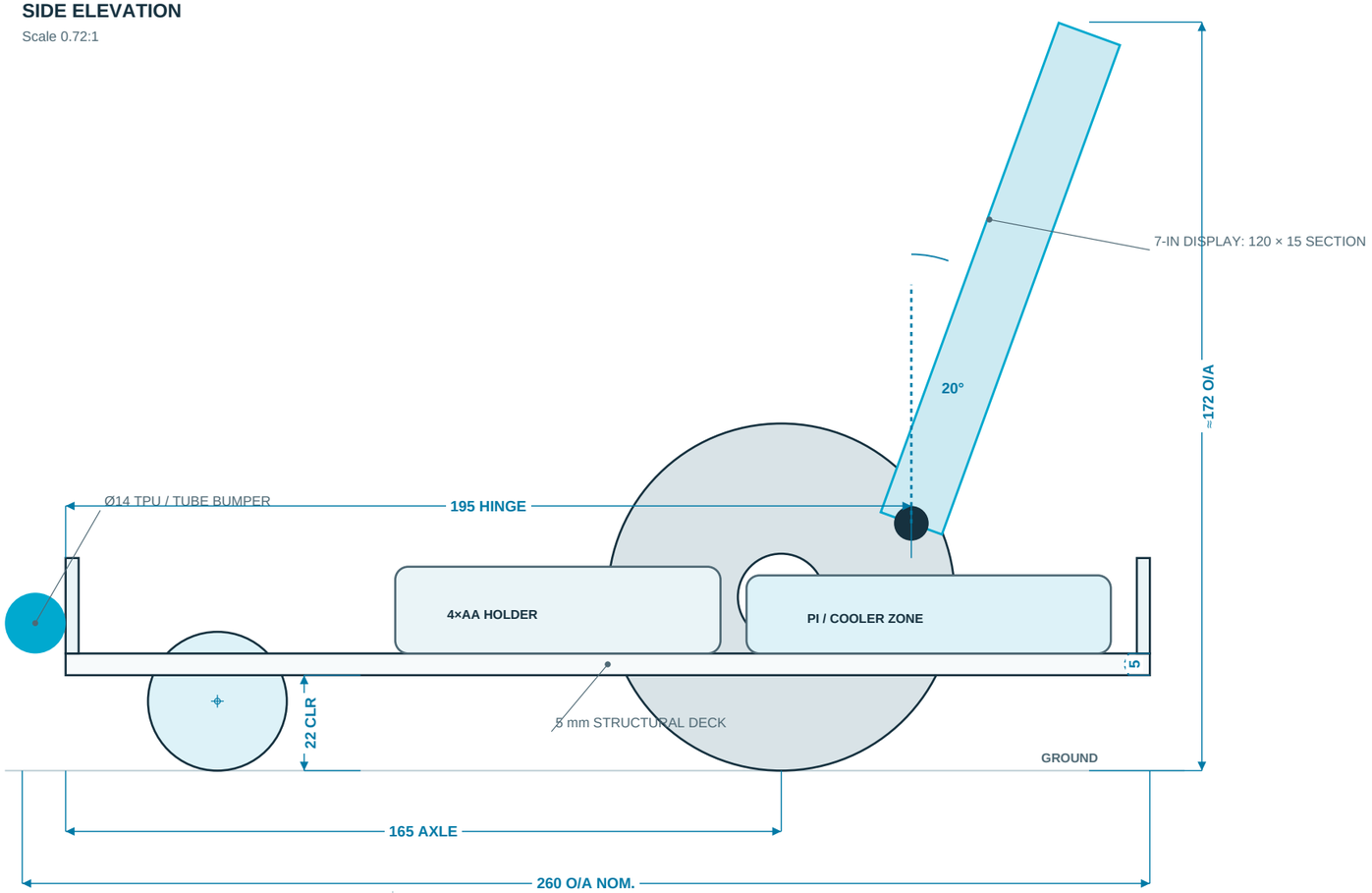
# ELEVATIONS & PRINT-BED SPLIT

Side/front geometry, clearances, display angle and optional two-piece joint



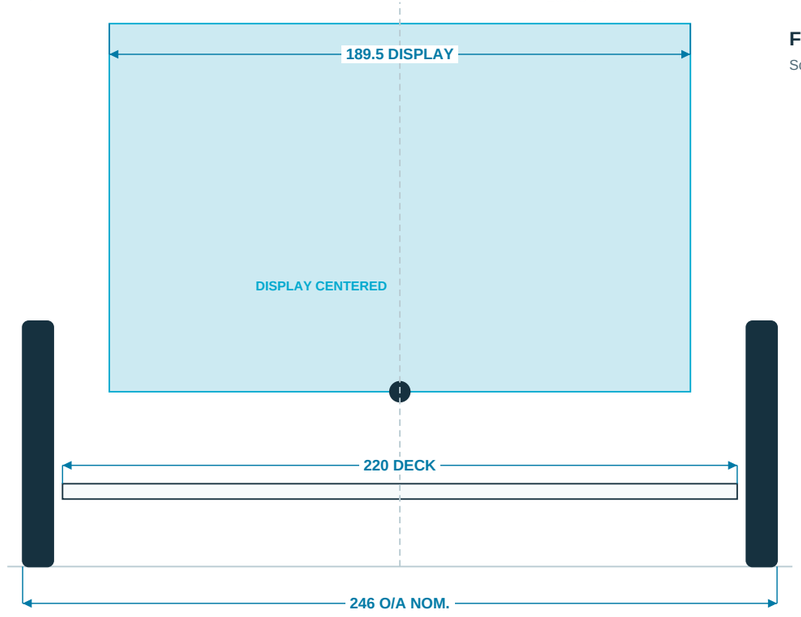
## SIDE ELEVATION

Scale 0.72:1



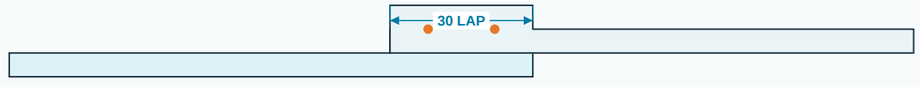

## FRONT ELEVATION

Scale 0.50:1




## OPTIONAL TWO-PIECE BASE — FOR BEDS <260 × 230

Split at X = 125; 30 mm overlapping lap joint




4× M4 through-bolts + washers; 2× Ø5 alignment dowels. Keep joint clear of battery and axle brackets.

| MUSE BOT BASE — MANUFACTURING SKETCH<br>Planning drawing; verify purchased hardware before production | DRAWING<br>MB-BASE-001 | REV<br>A | UNITS<br>mm | SCALE<br>AS NOTED |
|---|---|---|---|---|

# MANUFACTURING SCHEDULE & RELEASE CHECKS

Material, fit, mounting and verification notes for Rev A



## A. CONTROLLED GEOMETRY

| FEATURE | NOMINAL | DESIGN INTENT |
|---|---|---|
| Main deck | 250 × 220 × 5 | PA12-CF structural plate; R12 corners |
| Perimeter wall | 3 thick × 22 high | Local wall may be omitted at wheels and hinge |
| Ground clearance | 22 | Confirm after caster selection and wheel loading |
| Drive axle | X = 165; Z = 40 | Use adjustable motor-bracket slots |
| Caster center | X = 35; Y = 110 | Reserve Ø32 zone; slot pattern after purchase |
| Display hinge | X = 195; Z = 57 | 20° rearward from vertical; 189.5 span |
| Nominal envelope | 260 L × 246 W × 172 H | Includes bumper, wheels and raised display |

## B. HARDWARE ENVELOPES

| ITEM | REFERENCE SIZE | CAD / MOUNT TREATMENT |
|---|---|---|
| 7-in Touch Display 2 | 189.5 × 120 × 15 | Model exact outer envelope; verify rear boss pattern |
| Raspberry Pi 5 | 85 × 56 | Reserve 95 × 65 with active-cooler airflow |
| Arduino Uno R4 WiFi | 68.58 × 53.34 | Reserve 75 × 60; locate holes from physical board |
| MDD3A driver | 48.26 × 41.66 | Reserve 58 × 50; 4× Ø3 board holes |
| INA260 carrier | 22.9 × 22.8 × 2.7 | Nylon M2.5/M3 standoffs or adhesive carrier |
| VL53L1X carrier | 13 × 18 × 2 | Front-facing; removable two-screw carrier |
| 4×AA holder | ≈71 × 65 × 20 | Reserve 75 × 70; strap slots, not fixed holes |
| Gearmotor / wheel | Ø25 × 69 / Ø80 × 10 | 4 mm shaft; verify hub offset and bracket |

## C. HOLE & FEATURE STANDARD

| FEATURE | SIZE | USE |
|---|---|---|
| M3 clearance | Ø3.4 | Boards and light brackets |
| M4 clearance | Ø4.5 | Split joint and structural brackets |
| M3 adjustment slot | 4 × 12 | Motor/display bracket tuning |
| M4/caster slot | 5 × 15 | Caster height and tracking adjustment |
| Cable channel | 8 W × 6 D min. | Add R2 floor fillets and strain relief |

*Reference sizes derived from the handmade-bci-build-kit and manufacturer envelopes. Purchased hardware governs.*

## D. MATERIAL & PRINT PROCESS

### RECOMMENDED BUILD

Final structure: PA12-CF (carbon-fibre-filled nylon).
Prototype first in PETG before committing to PA12-CF.
TPU 95A for the bumper and vibration-isolation pads.
Use brass heat-set inserts; pilot diameters must follow the insert supplier.
Starting profile: 0.6 mm hardened nozzle, 5 walls, 6 top/bottom layers, 30–40% gyroid.
Dry PA12-CF and follow the filament maker's enclosure/nozzle guidance.
Use nylon standoffs or insulating washers beneath every circuit board.
Carbon-filled polymers can be electrically conductive: keep copper, solder joints and antennas isolated.

### TOLERANCE / DFM

General printed dimensions: ±0.5 mm unless noted.
Hole position after calibration: ±0.3 mm target.
Minimum wall: 3 mm. Structural deck: 5 mm.
Minimum inside fillet: R2; avoid sharp load-bearing corners.
Moving-part clearance: 0.5 mm per side minimum.
Print this A3 sheet at 100%; do not use 'fit to page'.

### SMALL-BED OPTION

If the printer bed is smaller than 260 × 230:
Split the deck at X = 125 with a 30 mm overlap.
Use 4× M4 through-bolts, washers and locknuts.
Add 2× Ø5 dowel features for repeatable alignment.
Keep the seam away from axle brackets and battery strap slots.

### RELEASE CHECKLIST — REQUIRED BEFORE FINAL PRINT

- ■ Place every purchased part on a 1:1 paper print.
- ■ Measure caster, brackets, wheel hub spacing and display bosses.
- ■ Confirm wheel clearance through full rotation and deck flex.
- ■ Check USB, power-switch, battery and SD-card service access.
- ■ Print a 20 mm corner/hole coupon in the final PA12-CF profile.
- ■ Update the CAD hole table; then freeze Rev B for manufacture.

| MUSE BOT BASE — MANUFACTURING SKETCH<br>Planning drawing; verify purchased hardware before production | DRAWING<br>MB-BASE-001 | REV<br>A | UNITS<br>mm | SCALE<br>NTS |
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